

UNIT 3

MIND-CONTROL TECHNOLOGIES



Connect to the topic

1. Look at the photo.
What do you think the device on the person's head is used for?
2. Do you think this device could allow someone to control objects with their mind? Why or why not?
3. What are the benefits of controlling devices with your mind instead of your hands?
4. If you could control any object with your mind, what would it be and why?

WARM-UP VIDEO

WATCH AND SPECULATE. Watch the video about the mind-reading technology and discuss the questions below.

1. What are some practical uses for mind-reading technology?
2. What does the speaker mean by "a world without limitations"?
3. Should neurotech companies have access to our brain data? Why or why not?
4. How could this technology change communication between people?
5. Do you think governments should regulate mind-control technologies? Why or why not?
6. Are we ready for a future where mind-control technologies are common?



UNIT 3.1 BRAIN-COMPUTER INTERFACES

READING AND VOCABULARY

Task 1. EXPLORE THE WORDS. Match the words to their definitions.

1. neurons	a) not involving the introduction of instruments into the body
2. neuroscience	b) understandable; clear in meaning
3. invasive	c) the scientific study of the nervous system
4. non-invasive	d) relating to the mental processes of perception, memory, judgment, and reasoning
5. cortex	e) a technique that trains individuals to control their brain activity through real-time displays of brain function
6. electroencephalography (EEG)	f) a technique for measuring brain activity by detecting blood flow changes
7. magnetic resonance imaging (MRI)	g) the brain's outer layer responsible for high-level functions like sensory perception and decision-making
8. neurofeedback	h) nerve cells that transmit signals in the brain
9. intelligible	i) involving the introduction of instruments or substances into the body
10. cognitive	j) a method used to record electrical activity of the brain through electrodes placed on the scalp

Task 2. WORK WITH WORDS. Fill in the gaps in the sentences using the words from previous exercise.

1. Many people use _____ to train their brains and control BCIs better.
2. When you think about moving your hand, _____ send signals that a BCI can detect.
3. _____ BCIs can be very accurate, but they need surgery, which is a serious decision.
4. The BCI reads signals from the _____, where your thoughts happen.
5. Doctors use _____ to see which brain areas are active with BCIs.
6. Researchers are working on _____ methods to make BCIs safer and easier for everyone.
7. Scientists use _____ to watch brain waves and improve BCI technology.
8. The BCI must change brain signals into _____ instructions for the device to work.
9. BCIs can help improve _____ functions in people with brain disorders, making their lives better.
10. New findings in _____ help create better ways to use BCIs in rehabilitation.

Task 3. READ FOR DETAILS. Read the text and complete the concept map using words from the text up to two words for each blank.

How a Brain-Computer Interface Works

Imagine transmitting signals directly to someone's brain that would allow them to see, hear, or feel specific sensory inputs. The reason brain-computer interface systems work is because of the way our brains function. Our brains are filled with neurons, individual nerve cells. Every time we think, move, feel, or remember something, neurons generate small electric signals. Some of this signal escapes, allowing scientists to detect neural activity and use it to direct external devices. It can also work in reverse. Researchers could figure out what signals are sent to the brain by the optic nerve when someone sees the color red and replicate those signals for a blind person to "see" without eyes.

One of the biggest challenges facing BCI researchers is the mechanics of the interface itself. The least invasive method is a set of electrodes – an electroencephalograph (EEG) – attached to the

scalp. However, the skull blocks a lot of the electrical signal. For a higher-resolution signal, scientists can implant electrodes directly into the brain, but this requires surgery and can cause scar tissue formation, which blocks signals.

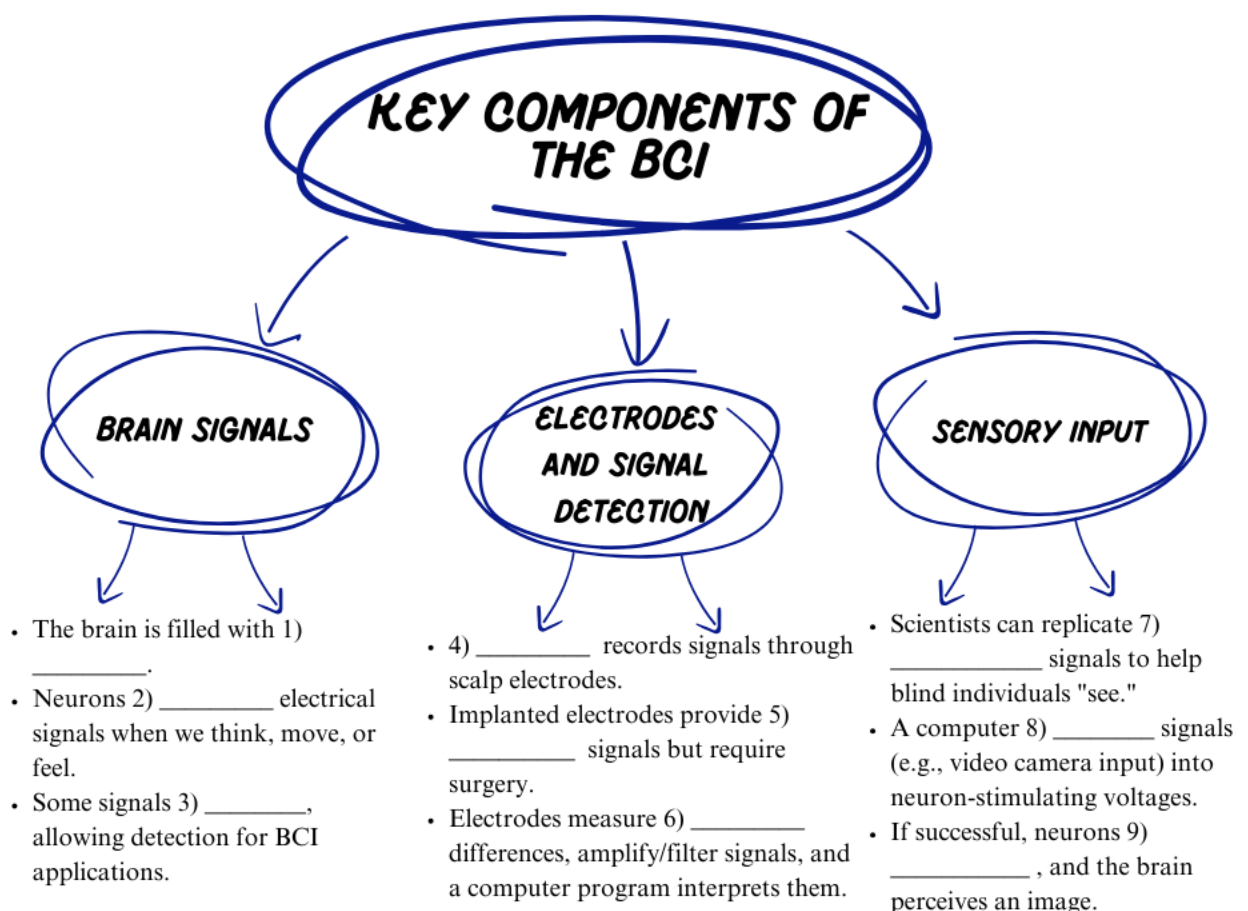
Regardless of electrode placement, the mechanism is the same. The electrodes measure voltage differences between neurons, amplify and filter the signal, and then interpret it through a computer program. In a sensory input BCI, a computer converts a signal, such as one from a video camera, into the voltages necessary to trigger neurons. If successful, the neurons fire, and the subject receives a visual image.

MRI provides high-resolution images of brain activity but can't be used as part of a permanent BCI. Instead, researchers use it to map where in the brain electrodes should be placed. For example, if attempting to implant electrodes that allow someone to control a robotic arm with their thoughts, researchers might first identify the brain area activated by arm movement.

One of the most exciting areas of BCI research is the development of devices controlled by thoughts. This technology has the potential to revolutionize the lives of disabled individuals. Early research used monkeys with implanted electrodes to control a robotic arm. Over time, researchers trained the monkeys to control the arm using only their brain signals. For people who cannot move their limbs, BCI software learns the signals associated with intended movement and translates them into action, such as closing a robotic hand.

A similar method allows users to manipulate a computer cursor, drawing circles, accessing programs, and even controlling a TV. With enough refinement, the technology could be expanded to allow users to "type" with their thoughts. Instead of a robotic hand, disabled users could have robotic braces attached to their own limbs, allowing them to move and interact with the environment.

(adapted from <https://computer.howstuffworks.com/brain-computer-interface.htm>)



Task 4. COMMUNICATE. Answer the questions below based on the information from the text.

1. Why do BCIs rely on detecting electrical signals from neurons?
2. What is the difference between EEG electrodes and implanted electrodes?
3. Why do implanted electrodes provide a better signal than EEG?
4. What challenges do scientists face when implanting electrodes in the brain?
5. What role does MRI play in BCI research?
6. How have monkeys contributed to BCI research?

Task 5. WORK WITH WORDS. Read the text and choose the correct variants for the missing words.

Signal Interpreters

In a brain-computer interface, signal interpreter programs are what make it possible to turn brain activity into real-world 1) _____. These computer programs take the tiny electrical signals from the brain and help translate them into digital instructions for external devices like robotic arms, wheelchairs, or computer 2) _____.

First, the system detects brain signals using 3) _____, these are usually placed on the scalp, though sometimes they're implanted directly in the brain. The raw signals that come through are often weak and noisy, so before anything else, the interpreter has to 4) _____ and amplify them. Only then can it begin analyzing the signals to recognize patterns linked to certain 5) _____ like when someone imagines moving their hand or focuses their attention on a screen.

When the system figures out what the brain is trying to do, it 6) _____ that into something the device can actually use like a movement or a click. These programs don't always get it right at first, but over time they usually adjust to how each person thinks or reacts. That's how they end up working better with practice. Without this kind of interpreter in place, it would be nearly impossible to turn 7) _____ into anything practical, especially for people who can't speak or move on their own.

- | | | | | |
|----|---------------|----------------|-------------|--------------|
| 1. | A) results | B) commands | C) systems | D) processes |
| 2. | A) wires | B) thoughts | C) cursors | D) circuits |
| 3. | A) electrodes | B) connections | C) tools | D) switches |
| 4. | A) clean | B) delete | C) build | D) filter |
| 5. | A) dreams | B) movements | C) opinions | D) language |
| 6. | A) reads | B) converts | C) connects | D) resets |
| 7. | A) ideas | B) codes | C) thoughts | D) messages |

Task 6. COMMUNICATE. Work with a partner. Answer the questions below.

1. What is the main function of signal interpreter programs in a brain-computer interface?
2. Why do the brain signals need to be filtered before being interpreted?
3. What kind of devices can these programs help control?
4. How does the signal interpreter learn to become more accurate over time?
5. What could happen if the signal interpreter misreads brain activity?

WATCHING

Task 7. COMMUNICATE. Work with a partner. Think of 3 questions you hope the video "Mind-reading AI" will answer. Check if they were answered after watching.

Task 8. WATCH FOR DETAILS. Scan the QR code and watch the video “Mind-reading AI” Choose the correct answer to the questions.



1. What are brain waves?
 - A. Physical waves moving in the brain
 - B. Electrical signals in the nervous system
 - C. Sounds that the brain makes
 - D. Light signals sent by neurons
2. What does an EEG do?
 - A. It measures the amount of blood in the brain
 - B. It creates brain waves
 - C. It measures brain waves
 - D. It counts the number of neurons in the brain
3. Why are brain waves difficult to read?
 - A. They are very weak
 - B. They disappear too quickly
 - C. They can only be seen with an MRI
 - D. They move too fast for sensors to detect
4. What is the main problem with MRI technology for reading thoughts?
 - A. It is too expensive
 - B. It is dangerous for brain activity
 - C. It is too slow
 - D. It is not small enough to carry around
5. What is the main advantage of invasive BCIs?
 - A. They do not need surgery
 - B. They can read brain signals more clearly
 - C. They are safer than MRI machines
 - D. They improve natural brain functions
6. What is Elon Musk’s Neuralink working on?
 - A. A way to increase memory power
 - B. A non-invasive brain interface
 - C. An invasive brain interface
 - D. An AI interface placed inside the brain
7. How does AI help brain-computer interfaces?
 - A. It makes the signals stronger
 - B. It translates brain signals into useful data
 - C. It creates new brain signals
 - D. It generates new thoughts in the brain
8. What is one possible future use of BCIs?
 - A. Controlling virtual objects with thoughts
 - B. Making people read faster
 - C. Stopping brain diseases forever
 - D. Replacing computers with human brains
9. What risk does BCI technology bring?
 - A. The brain could be controlled by a computer
 - B. The brain might stop working
 - C. People may lose their memories
 - D. Hackers may access people’s thoughts

Task 9. COMMUNICATE. Work in pairs. Discuss the questions below.

1. Would you feel comfortable using an invasive BCI? Why or why not?
2. Do you think BCIs should be available for everyday use, like VR headsets? Why?
3. If companies could read people's thoughts using BCIs, what problems might arise?
4. In what ways could BCIs impact mental health?
5. Do you think the potential for BCIs to be hacked is a big problem? Why?

LISTENING

Task 10. COMMUNICATE. You are going to listen to a dialogue where two friends are at a tech expo in a demo room testing a BCI device that allows people to move objects using only their minds. Before you start listening, discuss the questions with your partner:

1. Would you trust a device that can read your brain signals? Why or why not?
2. What do you think it would feel like to control something with your thoughts?

Task 11. LISTEN FOR DETAILS. Scan the QR code and listen to the audio "BCI Device", a conversation between Liam and Sasha testing a BCI headset. Answer the questions below.

1. What happens when Liam focuses on the blue block?
2. How does Sasha explain how the device works?
3. What does Liam compare the experience to?
4. How does Liam describe the feeling of moving something with his brain?
5. What does Sasha say the system needs to understand brain commands?
6. What kinds of technologies or objects do they say BCIs could control in the future?
7. What word does Liam use to describe moving objects with your mind?



SPEAKING

Task 12. CATEGORIZE. Read the explanations and categorize the following items. Write them in the correct column.

Invasive Vs. Non-invasive

Invasive BCIs are implanted directly into the brain through surgery. They offer very precise signals and are often used in medical cases, such as helping people with paralysis control robotic limbs or restore movement.

Non-invasive BCIs use devices placed outside the head, like EEG headsets, to read brain activity. These are commonly used in research, gaming, smart home systems, and other everyday technologies. They are safer and easier to use but may be less accurate.

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.

Risk of infection	Robotic arm controlled by brain signals	Used at home
Smart home control with headband	Neural implant to restore movement	Precise signal from brain
Brainwave-reading cap	High cost of surgery	Used in hospitals

Headset that tracks attention	No surgery needed	Brain surgery
EEG headset for gaming	VR game using thoughts	

Task 13. COMMUNICATE. Work in pairs or small groups.

1. What are the advantages and disadvantages of each type?
2. Which type would you prefer to use in daily life? Why?
3. Do you think invasive BCIs should only be used for medical purposes, or could they be useful in other areas as well?
4. How could BCIs impact relationships between people, considering their potential to read thoughts or intentions?

Task 14. COLLABORATE. Work in small groups. Discuss the situation below.

BCI for a Superpower

Imagine that you could use a BCI to give yourself a superpower. What superpower would you choose, and how would the BCI technology work to enhance that ability?

Present your idea to the class.

Task 15. COMMUNICATE. Role play the situation below.

Student A: You are an entrepreneur. Your company has developed a BCI device that can read people's thoughts and influence their decisions.

Student B: You are a journalist investigating its potential ethical implications.

The entrepreneur describes the device, the journalist asks tricky questions. You should also cover the questions below.

- Is it ethical to create technology that can read or influence thoughts?
- What regulations should be in place for these kinds of technologies?

Task 16. COLLABORATE. Work in pairs or small groups. Discuss a hypothetical scenario where someone could hack into a person's brain using BCI technology. Describe it and discuss the questions below.

What If...You Could Hack Someone's Brain?

1. How could this be used for both positive and negative purposes?
2. What risks and benefits are involved?
3. Could there ever be a situation where brain hacking might be acceptable?
4. What should be done to prevent abuse?

Task 17. COMMUNICATE. Work with a partner. Discuss the questions below.

1. Is it possible to make a completely hack-proof BCI?
2. What parts of our thoughts or memories should be protected?
3. Do we have a right to privacy in our thoughts?
4. Is it possible to create laws that protect people's thoughts from being accessed by hackers?
5. Should the brain be considered a private space, like personal data or conversations?

Task 18. COLLABORATE. Work in small groups. Read each dilemma and discuss possible solutions.

- A government wants to use BCI technology to control soldiers' movements and actions during combat. Is it ethical to use BCI in military settings?
- A scientist proposes using BCI to change or remove traumatic memories. Would it be right to offer this procedure to people in need, or does it cross an ethical line?

- A criminal is convicted of a crime, and the court decides to use BCI technology to access the criminal's thoughts and memories to verify guilt or innocence. Should BCI be used as a tool for legal decisions, or does it violate the person's right to privacy?
- BCI technology is used to help individuals overcome addiction by modifying brain signals associated with cravings. Should BCI be used to change someone's thoughts and impulses to treat addiction, or does it raise concerns about autonomy and free will?
- A government introduces a nationwide BCI program to monitor citizens' thoughts and emotions in real time to prevent criminal behavior and social unrest. While this might ensure greater safety, does this kind of surveillance violate human rights and freedom of thought?
- Entertainment companies want to use BCI technology to enhance viewers' experiences by controlling emotions during movies or games (e.g., making you feel more scared or excited). Is it ethical to manipulate people's emotions for commercial gain, or should there be restrictions on this?

Task 19. COLLABORATE. Work in small groups. Read the scenarios and discuss the future of BCI.

Future of BCI Technology

Brain-to-Brain Communication

Imagine being able to send a thought directly from your mind into someone else's – no typing, no speaking, just instant brain-to-brain exchange. It sounds efficient, maybe even exciting. But would it really improve how we understand one another, or could it actually create more confusion?

Upload and Download Thoughts

Scientists are exploring BCI systems that may allow people to upload their thoughts to a computer or even send them to another person. It's a powerful idea with lots of potential, especially in learning or teamwork. But what would happen if people began "sharing" ideas instead of learning them? Could this blur the line between collaboration and control?

Immortality

There's a theory that one day, people might upload their consciousness into a digital environment and exist there indefinitely. In a way, this would mean outliving the body, perhaps even living forever. But is that really the same as being alive? Would it change what we value in human experience?

Memory Transfer

BCI tools could also allow people to pass their memories to others, almost like sending a file. But memories are emotional and personal and would sharing them feel the same as experiencing them? And if people start collecting others' memories, how do we define what truly belongs to us?

Which future scenario would you like to see come true, and why?

Task 20. ANALYZE. Read the case study and discuss possible solutions to the problem.

One rehabilitation clinic in Germany started using a non-invasive BCI system to help stroke patients during therapy. Basically, patients wear a headset that picks up brain signals and helps them control a robotic arm. At first, it seemed to work quite well, people were showing progress, especially with motor control. But after a while, some problems started popping up. During sessions, a few patients experienced weird delays or sudden dropouts, like the arm just froze mid-move or didn't respond in time. This seemed to happen more in rooms with a lot of other electronics going on. The team working on the system suspects that the electrical noise from nearby devices is messing with the EEG signals, which makes sense. They're not planning to stop using it altogether as the early results were promising enough. But now they're trying to figure out how to make it more stable so it works better in those tricky environments.

What could they do in terms of design or setup to cut down the interference and help the BCI system run more smoothly in a busy medical setting?

LANGUAGE FOCUS

Task 21. STUDY AND ANALYZE. Look at the rule about participles, study in what situations they are used.

PARTICIPLES

Present Participle
playing, running, etc.

Past Participle
stayed, left, etc.

Perfect Participle
having left, etc.

Use	Example
to replace a relative clause	The signals controlling the robotic arm came from the patient's brain. (which were controlling) The device placed in the patient's brain helped them move again. (which was placed)
with prepositions and conjunctions	After testing the brain-computer interface, the scientists wrote a report. After having built the device, the team checked if it worked.
to explain the reason for something	Being interested in brain-computer interfaces, Sarah joined the research team. Having used similar technology before, Tom found the new system easy to understand.
to talk about actions happening at the same time	Watching the screen, the researcher saw how the brain signals changed.
to talk about actions happening in sequence	Having set up the device, the scientist started the experiment.
as an alternative passive form	Given a brain-computer interface, the patient could control a robot arm. (when / because he was given one)
as an alternative conditional form	Given the chance, I would try a brain-computer interface. (if I were given the chance)

Task 22. PRACTICE. Circle the correct word or phrase. If both are correct, circle both.

1. **After testing** / **having tested** the brain-computer interface, the researchers analyzed the data.
2. **Having been trained** / **Training** on large datasets, the AI could process brain signals accurately.
3. **Given** / **Giving** more funding, the team could develop a more advanced neural implant.
4. The researcher **analyzing** / **who was analyzing** the neural signals noticed a pattern.
5. **Having read** / **Reading** about brain-computer interfaces, Lisa decided to study neuroscience.
6. The patient **using** / **having used** the brain-computer interface successfully controlled a robotic arm.
7. **Not being** / **Not to be** an expert in AI, David asked for help with the algorithm.
8. **After decoded** / **having decoded** the neural signals, the computer translated them into commands.
9. **Shown** / **Having been shown** how to operate the interface, the participant started the test.
10. **Listening** / **Having listened** to the researcher's instructions, the team adjusted the experiment settings.

Task 23. PRACTICE. Write the verb in brackets in the correct form. Write up to three words in each gap.

1. The scientist hopes to improve brain-computer interfaces, _____ (give) more research funding.
2. After _____ (implant) the first BCI chip in a human, the company announced promising initial results.
3. The first brain-computer interface _____ (develop) in the 1970s was quite basic compared to modern systems.
4. Synchron's device was approved for human trials after _____ (demonstrate) its safety in animal studies.
5. _____ (study) neural signals for years, researchers at Blackrock Neurotech have improved BCI accuracy significantly.
6. _____ (track) brain activity in real time, the AI system converted thoughts into text.
7. _____ (train) on thousands of EEG recordings, the algorithm could predict motor intentions with high precision.
8. Is Neuralink the only company _____ (work) on high-bandwidth brain implants?
9. _____ (not want) to risk a failed experiment, the researchers double-checked all connections before testing.
10. _____ (stimulate) by electrical impulses, certain brain areas became more active.

Task 24. PRACTICE. Rewrite the sentences using participles.

e.g. The neural patterns which were detected during the experiment showed promising results.

The neural patterns, **detected** during the experiment, showed promising results.

1. The scientist connected the electrodes and started the brainwave recording.
2. After the engineers had programmed the AI, they ran the first simulation.
3. Because the signal was weak, the team recalibrated the sensors.
4. Lora adjusted the headgear before she activated the brain-computer interface.
5. The researcher was sitting at the desk and analyzing brain activity data.
6. The technician who is monitoring the brain signals is Dr. Patel.
7. Because the team had forgotten to charge the device, they had to postpone the test.
8. Daniel adjusted the electrodes and started the next phase of the experiment.
9. After the professor had explained the concept, he demonstrated a working prototype.
10. Because she had forgotten to update the software, Olivia had to restart the experiment.

UNIT 3.2 NEUROGADGETS

READING AND VOCABULARY

Task 25. EXPLORE THE WORDS. Match the words in bold with the correct definitions.

1. Many modern **brain-sensing** devices can track focus, relaxation, and stress levels by detecting electrical signals in the brain.
 2. A comfortable **headband** is often used in neurogadgets to keep sensors in place while monitoring brain activity.
 3. People use meditation and **mindfulness** techniques to improve their focus and reduce stress.
 4. Travelers often experience **jet lag** when their sleep schedule is affected by crossing multiple time zones.
 5. Some neurogadgets are designed to help people with **disabilities**, such as those with mobility or communication challenges.
 6. If your mind begins to **wander** during meditation, a neurogadget can provide real-time feedback to help you refocus.
 7. Many people **struggle** to concentrate for long periods, especially when surrounded by distractions.
 8. Good **sleep hygiene** can improve the effectiveness of sleep-tracking neurogadgets.
- a) a wearable strap that holds sensors in place for tracking brain activity.
 - b) a sudden problem in the body's sleep cycle due to traveling across different time zones.
 - c) the ability to stay fully present and aware of one's thoughts and surroundings.
 - d) a type of technology that detects brain activity and provides real-time feedback.
 - e) to find something difficult to do, often requiring extra effort.
 - f) a set of habits and routines that help improve sleep quality.
 - g) physical or cognitive conditions that make daily tasks more challenging.
 - h) to lose focus or drift away from a task unintentionally.

Task 26. WORK WITH WORDS. Complete the short text with the correct words from the previous task. You may need to change the form of the words.

Many neurogadgets rely on 1) _____ technology to detect brain signals and analyze mental states. These devices often come in the form of a 2) _____, which ensures the sensors remain in the correct position for accurate readings.

Practicing 3) _____ has become easier with smart neurogadgets that provide real-time guidance. However, some users find that their thoughts tend to 4) _____ during meditation, making it harder to stay focused. Special EEG-based devices help by alerting them when their attention drifts.

Frequent travelers often experience 5) _____, which disrupts their natural sleep cycles. To help restore a healthy routine, many people turn to neurogadgets designed to improve 6) _____. These devices track sleep patterns and suggest personalized adjustments.

Additionally, advanced brain-sensing technologies are opening new possibilities for individuals with 7) _____, enabling them to control devices using only their thoughts. Despite these benefits, some people still 8) _____ to maintain focus throughout the day which highlights the need for further innovation in cognitive enhancement.

Task 27. READ FOR MAIN IDEAS. Read the text about neurogadgets. Find out which device best matches the following statements. One of the statements doesn't relate to the text and should be marked "not given."

1. I track my sleep patterns using a brain-monitoring device.
2. I use a neurogadget to improve my meditation practice.
3. I use a neurogadget to help with physical exercise routines.

4. I use a neurogadget to control digital devices with my mind.
5. I rely on a brainwave headset to enhance my concentration.

The Future of Brain-Enhancing Technology

Alex, 29

Being a full-time software developer can get pretty intense, so I've been looking for ways to manage stress better. That's when I came across Muse, a brain-sensing headband designed to support meditation. It tracks your brain activity and sends feedback through an app. If your mind drifts off, you'll hear calming sounds that gently bring your focus back. I can't say it fixed everything overnight, but after using it regularly, I've noticed it's easier to stay calm and concentrate. Still, I think it's important not to depend too heavily on tech, traditional mindfulness techniques matter, too..

Lisa, 24

Studying at university can be a real challenge, especially when it comes to focusing for long stretches. I started using the NeuroSky MindWave, a lightweight EEG headset that monitors attention. It comes with training apps and even a few games, in some of them, I can move objects on the screen using only my thoughts. It's honestly kind of fun, and it does help me stay on task longer. That said, no gadget replaces good study habits. I try to balance both.

Mark, 35

Because I travel a lot for work, jet lag is something I've had to deal with often. Recently, I tried the Dreem Headband, which tracks sleep and uses gentle sounds to help improve deep sleep stages. The app gives me reports about my sleep cycles and offers suggestions that actually make sense. Since I started using it, I've been waking up feeling more rested. Still, I make sure to keep up basic sleep hygiene like dimming lights and staying off my phone before bed, because tech alone isn't a magic fix.

Sophie, 31

I've always had a soft spot for futuristic tech, so I couldn't resist trying NextMind. It's a small brain-sensing device that you place on the back of your head. It doesn't "read your thoughts," but it picks up brain activity and turns it into simple commands. For instance, I can move a cursor or choose a button on the screen just by focusing on it. It's not perfect and it's still in development but I think the potential is huge, especially for users with limited mobility.

Task 28. READ FOR DETAILS. Read the text again and choose the correct answers.

1. What is the main function of Muse?
 - A. Enhancing meditation and focus.
 - B. Helping people fall asleep faster.
 - C. Controlling digital devices with the mind.
 - D. Improving memory retention.
2. What does Alex say about using neurogadgets for mindfulness?
 - A. They should replace traditional meditation.
 - B. Meditation should also be practiced with and without them.
 - C. They are only useful for beginners.
 - D. They have no real effect on stress levels.
3. How does NeuroSky MindWave help Lisa?
 - A. It tracks her sleep cycles.
 - B. It improves her problem-solving skills.
 - C. It helps her stay focused while studying.
 - D. It assists her in controlling smart home devices.
4. What feature does the Dreem Headband offer?
 - A. It enhances concentration with brain-training games.

- B. It provides audio stimulation for sleep quality.
 - C. It allows users to meditate using sound feedback.
 - D. It translates thoughts into text on a screen.
5. What is special about NextMind?
 - A. It measures brain activity during sleep.
 - B. It provides relaxation techniques for stress management.
 - C. It allows users to control devices with brain signals.
 - D. It improves concentration levels while studying.
 6. What does Sophie believe about NextMind's potential?
 - A. It is a futuristic technology that may not be practical.
 - B. It is already widely used in everyday life.
 - C. It is mainly designed for meditation.
 - D. It could be useful for people with disabilities.
 7. What is a common theme in the speakers' experiences?
 - A. They all use neurogadgets for medical purposes.
 - B. They believe that neurogadgets can completely replace traditional methods.
 - C. They use neurotechnology to improve different aspects of their lives.
 - D. They all work in the field of neuroscience.
 8. What do the speakers say about the limitations of neurogadgets?
 - A. These devices are expensive and difficult to use.
 - B. They should be used alongside traditional methods, not as a replacement.
 - C. They can only benefit professionals in specific industries.
 - D. They do not provide any real benefits.

Task 29. COMMUNICATE. Work in pairs. Discuss the questions below.

1. Would you like to try a neurogadget from the text above? Which one and why?
2. Do you think brain-training devices like NeuroSky MindWave can really improve concentration?
3. How do you feel about using EEG-based devices to track sleep?
4. Could mind-controlled technology like NextMind become mainstream? Why or why not?
5. What other potential uses for neurogadgets do you think we will see in the future?

Task 30. WORK WITH WORDS. Read the text below about how companies market neurogadgets. Circle the correct word in each pair of options. Each pair has one word that fits best in the context of the sentence.

Marketing Neurogadgets

The neurotechnology industry is rapidly 1) **growing** / **growingly**, with companies focusing on improving user experience to attract customers. A key strategy in marketing neurogadgets is highlighting their 2) **accessibility** / **accessible**, ensuring that even non-experts feel confident using them.

To make their products more appealing, companies invest in 3) **consumer-friendly** / **consume-friendly** designs. Unlike earlier models, which were bulky and required complex setups, modern neurogadgets prioritize 4) **comfort** / **comfortable** and ease of use. Many devices now come with sleek 5) **headbands** / **headband**, wireless connections, and mobile app integration to enhance usability.

Another major trend is 6) **personalization** / **personal**, where neurogadgets adapt to individual users by analyzing their brain activity and providing 7) **customized** / **customizing** feedback. For example, meditation headsets adjust audio guidance based on real-time EEG data, ensuring a more 8) **effective** / **effectiveness** relaxation experience.

Companies also use various marketing strategies to increase **9) awareness / aware** and reach a wider audience. Social media influencers and tech reviewers play a major role in product **10) endorsements / endorsing**, helping brands build credibility. Additionally, many companies offer **11) trial / trying** periods, allowing potential customers to test the device before making a purchase.

The pricing of neurogadgets is another critical factor. While high-end models offer advanced features, many brands are now producing **12) affordable / affordability** alternatives to make neurotechnology more widely accessible. By focusing on **13) usability / use**, functionality, and market demand, companies continue to drive the rapid **14) expansion / expand** of neurogadgets in the consumer space.

Task 31. EXPLORE THE IDIOMS. Study the meaning and examples of IT related idioms. Then use them in sentences.

Mind over matter – using mental focus to overcome physical or technical challenges.

*The brain-computer interface proved **mind over matter** works with tech.*

Lose your marbles – to act irrationally, often due to tech overload.

*After hours tweaking the neurogadget, I thought I'd **lost my marbles**.*

Brainwave – a sudden clever idea or breakthrough.

*Her **brainwave** led to a new mind-control app feature.*

Wired differently – to think or function in an unconventional way.

*The neurotech team was **wired differently**, solving problems uniquely.*

Out of your mind – crazy or unbelievable, often about futuristic tech.

*A gadget that reads thoughts? Are you **out of your mind**?*

Pick someone's brain – to ask for advice or insights, often tech-related.

*I **picked the engineer's brain** about brain-interface latency.*

Mind the gap – to be cautious about a difference or risk.

***Mind the gap** between user expectations and neurotech reality.*

Fill in the gaps with the correct idiom from the list above. Each idiom is used only once. Sometimes you may need to change the form of the verb.

1. The designer's _____ added a voice-command feature to the neurogadget overnight.
2. "Control a drone with your thoughts? You must be _____!" the skeptic exclaimed.
3. After debugging the headset for hours, he felt like he'd _____, shouting at the screen.
4. We need to _____ between what users want and what the tech can actually do.
5. She _____ the lead scientist's _____ to understand how the brain signals were processed.
6. Their team was _____, coming up with solutions no one else had considered.
7. With practice, users showed that _____ could move a cursor just by thinking.

Task 32. COLLABORATE. Work in groups. "Sell" imaginary mind-control gadgets in a chaotic group auction, using idioms to hype them up. Each student invents a wild mind-control gadget (e.g., "Thought-Toaster" for breakfast ideas) and gets 10 fake "neuro-coins." Take turns as auctioneer, describing your gadget with at least two idioms. Others bid with coins, shouting why they want it. Highest bidder wins; discuss the priciest gadget after.

Task 33. WRITE. Write a one-day diary entry as a mind-control tech tester, but jumble the timeline, your partner should unscramble it. Use idioms in your diary.

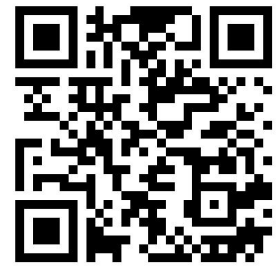
E.g. 12 p.m.: I lost my marbles when the helmet buzzed. 8 a.m.: A brainwave hit – adjust the signal!"

WATCHING

Task 34. EXPLORE THE WORDS. Study the sentences below and match the words in bold to their definitions.

1. The device uses **near-infrared light** pulses to improve brain performance and guide users through training exercises.
 2. Researchers analyzed **biomarkers** like attention, memory, and reaction speed to measure the effects of neurogadgets.
 3. The company published **white papers** showing scientific data on how their neurogadgets enhance brain function.
 4. Neurogadget introduced a **metric score** called Brain Years to help clinics assess cognitive abilities.
 5. The device helps users reach a **meditative state** by using audio feedback to guide their brain into relaxation.
 6. The neurogadget's Alpha Peak feature measures the **amplitude** of brainwaves in the 8-12 Hz range.
 7. Scientists study **Alpha brainwave frequency** because it plays a key role in relaxation and focus.
 8. The device tracks **blood flow** in the prefrontal cortex to measure concentration levels.
 9. Neurofeedback training increases **oxygenation** in the brain, which helps with focus and impulse control.
 10. The neurogadget displays **psychedelic patterns** of moving lights that may enhance relaxation.
 11. Exposure to bright light in the morning helps regulate the **circadian rhythm**, improving sleep and energy levels.
 12. Some studies show that flashing lights at 40 Hz may help prevent **cognitive impairment** linked to Alzheimer's disease.
 13. The neurogadget helped the reviewer understand when he was in a **flow state**, making work easier and more productive.
 14. One major **hurdle** for neurogadgets is getting people to wear them in public without feeling uncomfortable.
-
- a) signs in the body or brain that show how well it is working.
 - b) a deep, relaxed mental condition often reached through mindfulness.
 - c) the movement of blood through the body
 - d) colorful, shifting light visuals often used for relaxation.
 - e) reports with research and scientific data.
 - f) the strength or height of brainwave signals.
 - g) problems with thinking, memory, or focus.
 - h) a difficulty or challenge that must be overcome.
 - i) the body's natural sleep-wake cycle.
 - j) a number used to measure something e.g. brain performance.
 - k) a mental state of deep focus and effortless work.
 - l) the process of supplying oxygen to body tissues
 - m) a type of light used in brain devices to improve performance.
 - n) a type of brain activity linked to relaxation and focus.

Task 35. WATCH FOR DETAILS. Scan the QR code and watch the video “Neurogadgets”. Choose one device as the correct answer to the question.



Which device ...

1. has the largest guided meditation library?
2. uses near-infrared light instead of sensors to track blood flow in the brain?
3. looks like regular headphones but has brain-tracking features?
4. has a LED panel with psychedelic light patterns?
5. has a feature called Brain Years to measure brain health?
6. tracks a brainwave feature called Alpha Peak?
7. lets users move a ball on a screen using focus?
8. tracks focus levels throughout the day to understand work habits?
9. costs \$1700 and has a special version for clinics?
10. is the easiest to use because it doesn't need careful placement?

Sensei
Muse
Mendi
Neurovizr
Neurable

Task 36. WATCH FOR DETAILS. Decide if the sentences below are True or False.

1. The Muse headband helps users enter a meditative state by tracking their brainwaves.	True / False
2. Flashing lights at 40 Hz may help activate brain cells linked to memory and learning.	True / False
3. The NeuroVizr device requires EEG sensors to function properly.	True / False
4. NeuroVizr uses sound waves to measure cognitive performance.	True / False
5. The Sensei device creates a metric score called Brain Years to track cognitive performance.	True / False
6. The Muse S version includes sleep tracking and audio tracks to help with sleep.	True / False
7. The Neurable headphones help track physical exercise performance through AI-powered apps.	True / False
8. The Sensei device provides real-time reaction time testing to measure brain performance.	True / False

Task 37. COMMUNICATE. Work in pairs. Discuss the questions below.

1. Would you use any of these neurogadgets? Why or why not?
2. Which device do you think has the biggest impact on mental health?
3. How could neurogadgets like Muse and Sensei help students improve their focus?
4. Do you think neurogadgets will become as common as smart watches in the future?
5. What are some possible risks or disadvantages of using neurogadgets?

Task 38. COLLABORATE. Work in groups. Imagine you work for a company selling neurogadgets. Each group will choose one device from the video and prepare a short sales pitch convincing the class to buy it. Include:

- What the device does
- Who should buy it
- Why it is better than other options

Each group presents their pitch, and the class votes on the most convincing one.

LISTENING

Task 39. LISTEN FOR MAIN IDEA. Scan the QR code and listen to the dialogue “Muse Neurogadget” where two friends are talking about a neurogadget. Answer the questions below.

1. What is the main purpose of the Muse device?
2. What kind of feedback does the user get from the headband during meditation?
3. What are two features mentioned in the newer version of the Muse?



Task 40. LISTEN FOR DETAILS. Listen to the dialogue again and fill in the missing words.

1. Muse is a _____ headband that tracks brain activity.
2. You hear peaceful weather sounds if your mind is _____.
3. The sounds change to _____ or storm noises when your brain is distracted.
4. The Muse app shows how long you stayed calm and how often your mind _____.
5. The newer version can also track heart rate, _____, and movement.
6. One of the speakers uses it before _____ to relax.

SPEAKING

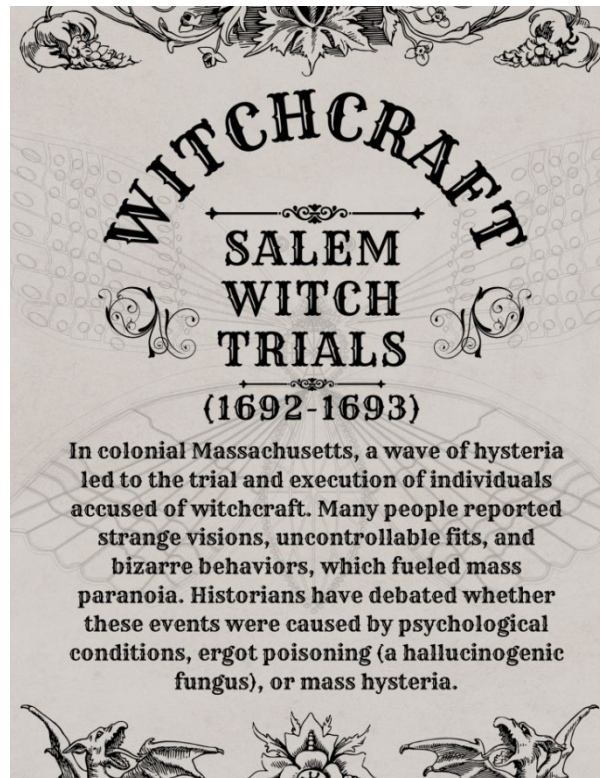
Task 41. COMMUNICATE. Work in pairs. Student A is a person interested in buying a neurogadget, and Student B is a tech expert who explains how it works. Read the short descriptions of OpenBCI Headsets and Emotiv Insight. Role play a conversation using the prompts provided, change roles. Try to ask follow-up questions and give detailed answers.

OpenBCI Headsets	Emotiv Insight
OpenBCI headsets are open-source EEG devices built for people who want to explore the brain in a hands-on way. Often used in brain-computer interface research, neurofeedback training, or experimental setups, these headsets provide high-quality recordings of brain signals. They give users a lot of flexibility, especially for building their own software or hardware add-ons. Still, these headsets aren't perfect, some technical know-how is definitely needed.	Emotiv Insight is a consumer-friendly device. It's a sleek-looking EEG headset, wireless too, and made to be much easier to handle. It reads brain activity in real time and gives you feedback like how focused or calm you are through its app. Emotiv Insight keeps things simple, so users don't have to interpret raw data themselves. Most people use it for meditation or boosting attention without any technical background.
Prompts	Prompts
Student A: Ask about what the OpenBCI headset does and who it is designed for. Student B: Explain its functions. Student A: Ask if it can be used for everyday tasks like meditation or gaming. Student B: Explain that it is more complex and not as user-friendly as commercial neurogadgets. Student A: Ask about the advantages of an open-source EEG device. Student B: Highlight the flexibility, customization options, and potential for innovation. Student A: Thank the expert and decide whether or not to consider the purchase.	Student A: Ask about how the Emotiv Insight headset works. Student B: Explain what it monitors and provides. Student A: Ask if it is difficult to set up and use. Student B: Explain that it is wireless and comes with a user-friendly app for tracking cognitive performance. Student A: Ask about practical uses. Student B: Explain how it helps. Student A: Express interest and ask about the price or where to buy it. Student B: Provide details on availability and mention any similar alternatives

Task 42. COLLABORATE. Work in small groups to discuss how neurogadgets could have helped reveal the real causes behind the Salem Witch Trials. Consider:

- How EEG-based headsets like Emotiv Insight might have helped understand whether the accused were affected by stress, illness, or neurological conditions.
- Whether neurofeedback tools could have helped people manage fear and reduce the panic spreading through the town.
- How sleep-tracking devices such as Dreem could have shown if exhaustion played a role in hallucinations or irrational behavior.
- How modern brain-sensing technology might help researchers today study mass panic or fear-based decisions.

Then discuss the questions below.



1. Could neurogadgets have helped prove whether the accused were suffering from real neurological conditions or external factors?
2. How might brainwave analysis have influenced the trials and prevented wrongful executions?
3. Could neurofeedback have been used to calm hysteria and reduce paranoia?
4. How might similar technology be used today to study or prevent mass hysteria in society?

Task 43. COLLABORATE. Work in small groups. Explore how neurogadgets could help prevent workplace stress and burnout.

Workplace stress and burnout are increasing concerns in modern society, particularly in high-pressure environments like technology, healthcare, and finance. These conditions can affect mental health, productivity, and overall well-being.

Think about the following aspects:

- EEG devices for early detection of stress patterns.
- Neurofeedback technology for emotional regulation.
- Sleep trackers to improve sleep quality and reduce stress.
- Brainwave analysis to enhance focus and productivity.

Discuss the questions:

1. How could neurogadgets identify stress before it leads to burnout?
2. In what ways can neurofeedback improve emotional regulation at work?
3. Could sleep-tracking gadgets help reduce stress in employees?
4. How might brainwave analysis create a healthier work environment?

Task 44. COLLABORATE. Play a game in small groups. One student picks a gadget card and keeps it secret. The others ask yes/no questions to figure out what it is. The student can only answer with "Yes," "No," or "I don't know."

Guess the Gadget

NeuroHalo (meditation headset)	DreamBand (REM sleep controller)
FocusRing (attention alert wearable)	MindZap (stress relief wristband)
SleepSync (EEG smart pillow)	ThoughtLink (brain-to-text device)
AlphaPatch (mood-stabilizing neural sticker)	NeuroBuddy (voice-controlled concentration tracker)

Task 45. COLLABORATE. Divide into two teams to debate the use of neurogadgets for mental health treatment.

Some people argue that neurogadgets should be widely used to treat mental health conditions, while others believe that traditional methods are still the best option. What do you think? Neurogadgets, such as EEG-based devices and neurofeedback tools, promise to help manage stress, anxiety, and depression. But are they reliable and accessible for everyone?

Should we use neurogadgets for mental health treatment?

YES	NO
1. Neurogadgets can offer real-time monitoring of brain activity, allowing for personalized treatment.	1. Traditional treatments like therapy and medication have been proven effective for many years, while neurogadgets are still new and untested in large populations.
2.	2.

Task 46. ANALYZE. Read the case study and discuss possible solutions to the problem.

A health tech company recently introduced a new neurogadget – a smart headband that's supposed to track brain activity while you sleep. It works by using EEG sensors to monitor your sleep stages, and it can also adjust things in the room automatically, like lighting or temperature, based on your brain activity. Some early users said it helped them sleep better, which was a good sign. But not everyone had the same experience. A number of people complained about skin irritation, and some said the headband didn't stay in place if they moved around during the night. There were also reports that the data didn't always make sense or seemed inconsistent. From what the company has said, they think it might have to do with how the sensors are placed or maybe the type of materials being used.

What kinds of design tweaks or technical changes could help make the device more comfortable and reliable for users during sleep?

LANGUAGE FOCUS

Task 47. STUDY AND ANALYZE. Look at the rule about linking words, study in what situations they are used.

LINKING WORDS

Linking words show the logical relationship between sentences or parts of a sentence.

Use	Example
Positive Addition and, both ... and, too, besides (this/that), moreover, what is more, in addition (to), also, as well as (this/that), furthermore	The new neurogadget is both compact and highly efficient.
Negative Addition neither ... nor, nor, neither, either	Neither the headset nor the implant required external power sources.
Contrast but, although, in spite of, despite, while, whereas, even though, on the other hand, however, yet, still	The device is powerful but still lightweight enough to wear comfortably.
Giving Examples such as, like, for example, for instance, especially, in particular	Many companies, such as Neuralink and Synchron, are developing advanced neurogadgets.
Cause/Reason as, because, because of, since, for this reason, due to, so, as a result (of)	Because of its high accuracy, the brain-computer interface became a breakthrough in assistive technology.
Condition if, whether, only if, in case of, in case, provided (that), providing (that), unless, as/so long as, otherwise, or (else), on condition (that)	The device will function properly only if the electrodes are correctly positioned.
Purpose to, so that, so as (not) to, in order (not) to, in order that, in case	The neurogadget was designed to detect brain signals so that users could control smart devices hands-free.
Effect/Result such/so ... that, so, consequently, as a result, therefore, for this reason	The neural implant was so advanced that it allowed paralyzed patients to type with their thoughts.
Time when, whenever, as, as soon as, while, before, until/till, after, since	As soon as the signal was detected, the robotic arm moved accordingly.
Place where, wherever	Neurogadgets can be used wherever brain activity needs to be monitored in real time.
Exception except (for), apart from	Apart from minor calibration issues, the device worked flawlessly.
Relatives who, whom, whose, which, what, that	The scientist who developed the neurogadget won an innovation award.
Listing Points/Events To begin: initially, first, at first, firstly, to start/begin with, first of all To continue: secondly, after this/that, second, afterwards, then, next To conclude: finally, lastly, in the end, at last, eventually	First of all , the researchers tested the device on volunteers. Next , they analyzed how accurately the gadget translated brain signals into commands. Finally , the neurogadget was ready for clinical trials.

Summarizing:

in conclusion, in summary, to sum up, on the whole, all in all, altogether, in short

To sum up, neurogadgets have the potential to revolutionize human-machine interaction.

Task 48. PRACTICE. Choose the correct linking word/phrase.

1. Brain-computer interfaces are improving **but/as** they still have some problems.
2. You can use the neurogadget at home **as long as/apart from** you follow the instructions.
3. **Afterwards/In conclusion**, the researchers showed their new brain implant.
4. The team improved the device **in order to/in case** make it work better.
5. They had to fix the software **because/in short** it wasn't reading brain signals well.
6. The new headset is easy to use. **However/In addition to**, it needs charging often.
7. The user thought about moving the cursor, **so/for example** the neurogadget followed the command.
8. **Although/All in all** the brainwave headset was expensive, it worked very well.
9. The AI assistant turned on **when/then** the user focused on the screen.
10. Some companies, **such as/moreover** Neuralink and Synchron, make brain-controlled devices.
11. The new implant worked well, **whereas/apart from** the older version, which had problems.
12. The engineer was changing the settings **while/despite** the patient was testing the device.

Task 49. PRACTICE. Join the sentences using the word(s) in brackets.

1. The brain implant needed fixing. The technician ran tests. (so)
2. The headset tracks brain activity. It also gives real-time feedback. (what is more)
3. The signal was weak. The device still worked. (despite)
4. The researcher was watching the brainwaves. The AI was analyzing the data. (while)
5. One headset uses dry sensors. The other needs gel. (whereas)
6. The neural interface read the user's thoughts. The robotic arm moved. (as soon as)
7. The engineer adjusted the settings. He tested the device again. (then)
8. The team finished testing. They shared their results. (after that)
9. The neurogadget helps disabled people. It also improves gaming. (as well as)
10. The company didn't use wires. It also didn't need headbands. (neither)

Task 50. PRACTICE. Fill in the correct word or phrase from the box. There are extra variants you do not need to use.

eventually ▪ first of all ▪ still ▪ unlike ▪ as a result ▪ except ▪ to sum up ▪ what is more ▪ because of this ▪ neither

The Stentrode is a groundbreaking brain-computer interface designed to help people with severe physical disabilities control technology using their thoughts. 1) _____ traditional brain implants, it is inserted through blood vessels, making the procedure much less invasive.

2) _____, the Stentrode records brain signals from the motor cortex and transmits them to a computer. 3) _____, users can operate devices like computers or prosthetic limbs just by thinking. This technology could significantly improve independence and quality of life for people with conditions like ALS or spinal cord injuries.

4) _____, the Stentrode does not require open-brain surgery, reducing risks and making it more accessible. 5) _____, researchers believe it could become a widely used tool for assistive communication and mobility.

6) _____, the Stentrode is a promising step toward practical mind-controlled devices, and 7) _____, as research advances, it may open even more possibilities in neurotechnology.

UNIT 3.3 MIND-CONTROL TECH AROUND US

READING AND VOCABULARY

Task 51. EXPLORE THE WORDS. Read this short introduction to mind-controlled technology and find words that match the meanings below.

Mind-controlled technology, often referred to as brain-computer interface, is no longer just science fiction. It is already being used in a variety of areas, including healthcare and entertainment. These systems let people interact with digital devices simply by using their brain activity, without needing to move their body.

In healthcare, one of the more exciting applications of BCI technology has been the creation of neuroprostheses – brain-controlled tools that give people back certain functions they’ve lost. These aren’t just experimental gadgets; they’re real assistive technologies already being used in practice. For someone who’s lost the ability to move or speak due to a stroke, accident, or illness, these systems can offer a new kind of independence. In fact, many users have been able to control wheelchairs, robotic arms, or even digital speech devices simply by focusing their thoughts. It’s not magic but it’s really close.

To get the best results, we often need to calibrate these technologies for each user. This is important because the device has to learn how to read that person’s unique brain patterns. This process makes the system more accurate over time. And when it comes to implants or anything worn inside the body, materials must be biocompatible, safe for long-term contact with tissue. If not, the body may react badly, possibly leading to adverse effects like swelling or rejection of the device.

But healthcare isn’t the only place BCI is making an impact. In gaming and virtual reality, there are headsets which let players perform actions by focusing their thoughts. Smart home applications are also growing. Imagine switching off lights or locking your front door without lifting a finger, simply by using your mind.

1. _____ advanced brain-controlled devices that help restore lost functions like movement or communication.
2. _____ type of technology designed to support individuals with disabilities in daily tasks.
3. _____ mobility devices to help individuals move independently.
4. _____ to adjust a device so it accurately interprets brain signals.
5. _____ safe to use inside the human body without causing harm.
6. _____ harmful effects that may result from using a medical device.

Task 52. COLLABORATE. Work in pairs. Choose the best number to complete the sentences about mind-controlled technology. Discuss your answers and check with your teacher.

1. By 2030, the global BCI market is expected to be worth **\$5 billion / \$8 billion / \$12 billion**.
2. Mind-controlled gaming systems can detect brain activity in **1 second / 5 seconds / 10 seconds**.
3. Neural implants have successfully restored movement in **10% / 40% / 80%** of paralyzed patients in recent clinical trials.
4. In the future, BCI-controlled smart homes could reduce energy consumption by **5% / 15% / 30%**.
5. More than **50,000 / 100,000 / 500,000** people worldwide are already using neuroprosthetic devices.

Which fact about mind-controlled technology surprises you the most? Explain your answer.

Task 53. EXPLORE THE WORDS. Complete the following definitions using phrases from the box.

a) that restore	b) mental exhaustion	c) robotic frame
d) response based on	e) vibrations or pressure	f) instructions using
g) able to adapt	h) smart controls	i) through brain activity

1. **haptic feedback** (phr) – _____ that simulate touch.
2. **neuroprosthetics** (n) – brain-controlled devices _____ movement or communication.
3. **exoskeleton** (n) – a wearable _____ that enhances movement.
4. **cognitive fatigue detection** (phr) – a system that identifies _____.
5. **mental effort calibration** (phr) – adjusting device _____ cognitive load.
6. **attention tracking** (phr) – monitoring focus levels _____.
7. **mood-based** (adj) – being _____ to emotional states.
8. **neural interface hubs** (phr) – systems that connect BCI devices with _____.
9. **EEG-driven commands** (phr) – thought-based _____ brain signals.

Task 54. READ FOR DETAILS. Work in A/B/C groups. Each group will read about a different real-world application of mind-controlled technology.

1. First, look through your information and use three words from the word box to complete the text.
2. Prepare a mini presentation about your topic for the other students in your group.
3. Listen to the other presentations and ask questions.
4. Read through the other two texts and add the missing words.

neural interface hubs ▪ haptic feedback ▪ neuroprosthetics ▪ mood-based ▪
cognitive fatigue detection ▪ EEG-driven commands ▪ exoskeleton ▪
mental effort calibration ▪ attention tracking ▪

Group A

Mind-Controlled Mobility Solutions

People with severe mobility impairments can now use 1) _____ to control wheelchairs and robotic limbs. These systems use 2) _____ to translate brain activity into movement, allowing users to navigate their environment independently. Some of these technologies involve 3) _____, which provide external support and assist users in regaining mobility through BCI-driven motion control.

Group B

Thought-Controlled Smart Homes

The integration of 4) _____ in homes allows users to operate lights, appliances, and security systems using only their thoughts. These systems rely on 5) _____ to adjust responses based on the user's cognitive load. Some smart home systems also incorporate 6) _____ adjustments, where lighting and temperature settings adapt based on the user's emotional state.

Group C

BCI in Gaming and Entertainment

Brain-controlled gaming is an emerging field where players can interact with virtual worlds using 7) _____. These systems analyze 8) _____ to assess how long a player can maintain focus, allowing for adaptive difficulty levels. Some neurogaming companies also

incorporate 9) _____, providing physical sensations that enhance immersion and make virtual experiences feel more realistic.

Task 55. COLLABORATE. The success of mind-controlled technology depends on responsible development and ethical use. Work in pairs or small groups to match each piece of advice to one or more of the situations you read about in texts A-C.

1. Make neuroprosthetics and exoskeletons more affordable and available.
2. Protect neural data in smart home systems from hacking.
3. Allow users to adjust mental effort settings to avoid fatigue.
4. Set clear rules on using attention tracking in gaming and education.
5. Ensure haptic feedback is comfortable and safe.
6. Teach users how EEG-driven commands work and their risks.
7. Help people with disabilities adapt to neural interface technology.
8. Discuss privacy concerns about smart neural interfaces.

Think of one more piece of advice for each situation. Use words from the lesson.

Task 56. WORK WITH WORDS. Fill in the gaps with the words from the box below. There are two extra words you do not need to use.

Mind-Controlled Exoskeletons and Neuroprosthetics

brain signals ▪ electrodes ▪ mobility ▪ spinal cord ▪ controller ▪
artificial limb ▪ rehabilitation ▪ battery ▪ wearable ▪ feedback ▪
implant ▪ grip strength ▪ amputees

Modern medical technologies such as 1) _____ and exoskeletons are changing the lives of people with physical disabilities. These systems are often 2) _____ devices that detect neural activity or muscle movement and help restore 3) _____ to paralyzed limbs.

In many systems, 4) _____ are used to detect 5) _____ from the brain or nerves, which are then interpreted by a central 6) _____. This allows users to move a robotic limb or walk with the help of an exoskeleton simply by thinking about the movement.

Some neuroprosthetics include sensors that provide 7) _____ to the user, helping them control their 8) _____ when holding or manipulating objects. These advances are especially useful in 9) _____ centers, where patients learn to use their devices over time and improve their independence.

From helping stroke patients walk again to enabling 10) _____ to control a robotic arm, these technologies demonstrate how brain-computer interfaces can reconnect the mind and body through innovation.

WATCHING

Task 57. COMMUNICATE. Work in pairs. Before watching the video, think about what might be discussed based on its title "Brain Chips." Answer the following questions. After watching, check your predictions.

1. What do you think the video will say about brain chips?
2. Who might benefit from brain chips?
3. How do you think brain chips connect to artificial intelligence?
4. What risks do you think brain chips might have?
5. Do you think brain chips are available for everyone now? Why or why not?

Task 58. WATCH FOR DETAILS. Scan the QR code and watch the video “Brain Chips”. Choose the correct answers to the questions.



1. Who was the second patient implanted with Neuralink?
 - A. A person with ALS
 - B. A person with a spinal cord injury
 - C. A scientist working with Elon Musk
2. What does Neuralink allow patients to do?
 - A. Control technology with their mind
 - B. Walk again after a spinal injury
 - C. Improve their memory
3. What was Nolan Arbaugh’s reaction to receiving the Neuralink implant?
 - A. He was disappointed with the results
 - B. He was afraid of using new technology
 - C. He was excited to be part of an important step in helping people with paralysis
4. According to Elon Musk, why might people without disabilities want a brain chip?
 - A. To gain superhuman abilities
 - B. To cure diseases
 - C. To increase their lifespan
5. What other company is working on brain implants besides Neuralink?
 - A. OpenAI
 - B. Tesla
 - C. Synchron
6. What did the 64-year-old patient with ALS do using a brain implant?
 - A. He sent text messages and played games
 - B. He walked with robotic legs
 - C. He talked without speaking
7. What device did the ALS patient use to interact with technology?
 - A. An AR headset
 - B. A smartwatch
 - C. A mobile phone
8. What does Elon Musk believe about the future of brain chips?
 - A. They will only be for medical use
 - B. They might be used by everyone to compete AI
 - C. They will replace all existing computers

Task 59. COMMUNICATE. Work in pairs. One student describes an optimistic future for the patient after receiving the brain implant, while the other describes a pessimistic future. Include details about his/her daily life, abilities, career opportunities, integration into digital life, technical failures, ethical concerns, or long-term side effects. Decide which scenario seemed more realistic?

SPEAKING

Task 60. COLLABORATE. Work in groups. Read the statements below about brain chips and decide whether they are FACTS (proven information) or SPECULATION (opinions or predictions). Then answer the questions below.

Fact vs. Speculation

1. There is no risk of failure or complications with brain chip technology.
2. In the future, brain chips might be used to enhance memory and learning speed.
3. Brain implants have already cured paralysis in multiple patients.
4. Some brain implants can help patients control robotic limbs using their thoughts.

5. Brain chips will eventually allow people to communicate telepathically.
6. The human brain can generate electrical signals that can be detected and interpreted by brain-computer interfaces.
7. In the next decade, most jobs will require people to have brain chips for faster data processing.
8. Some brain implants are already being tested to help patients with depression and anxiety.
9. Brain chips will completely replace traditional education, as people will be able to download knowledge directly into their brains.
10. Brain implants can help restore vision in certain cases of blindness.
11. Governments will soon require all citizens to get brain implants for identification and monitoring.

- How did you decide if something was a fact or speculation?
- Which speculative statements do you think are most likely to become true?

Task 61. COMMUNICATE. Work in pairs. Role play a conversation where one student describes strange things happening to a mind-control device, and the other student will act as a tech support specialist trying to diagnose the problem. Take turns switching roles.

Situation 1

Whenever I try to focus on a report at work using my mind-control headset, it keeps showing me random images of beaches, mountains, and cities, which distract me from my task. It's getting frustrating because I can't seem to get back to my work.

Tech Support Response: Diagnose the issue (possible settings adjustment, system bug, etc.)

Situation 2

In my classroom, my students are using mind-control devices to stay focused during lessons, but lately, whenever a student tries to concentrate, the device starts to play calming music that makes them sleepy. I need help figuring out why this keeps happening and how to fix it.

Tech Support Response: Diagnose the issue (device malfunction, interference with concentration settings, etc.)

Situation 3

I've been using a VR headset with mind-control features to play a game, but during one level, I started experiencing vivid, unsettling dreams that match the game's plot. The dreams feel so real that I can't tell if I'm still playing or actually dreaming. Is this a problem with the mind-control tech?

Tech Support Response: Diagnose the issue (possible malfunction, emotional overload, etc.)

Situation 4

"I've been using a mind-control productivity device to help me manage my tasks and stay motivated, but it now keeps suggesting random tasks that I never programmed into the system. Yesterday, it told me to 'declutter the attic' even though I don't have an attic. Is this a glitch?"

Tech Support Response: Diagnose the issue (faulty AI suggestions, software update needed, etc.)

Situation 5

I've been using mind-control technology for my VR gaming sessions, but recently, I keep hearing voices in my head during the game, even when I'm not talking. It's like the game is tapping into my subconscious thoughts. Is that a feature or a bug?

Tech Support Response: Diagnose the issue (mental feedback loop, settings interference, etc.)

Situation 6

I'm using a mind-control device to remotely operate machinery at work, but when I try to make small adjustments, the machine starts acting unpredictably and moving in wrong directions. Could this be due to a fault in the connection?

Tech Support Response: Diagnose the issue (signal interference, device calibration, etc.)

Situation 7

I'm using a mind-control headset for wellness purposes, but every time I use it, I start to experience vivid memories of things I thought I had forgotten. Should the device be triggering these memories, or is it malfunctioning?

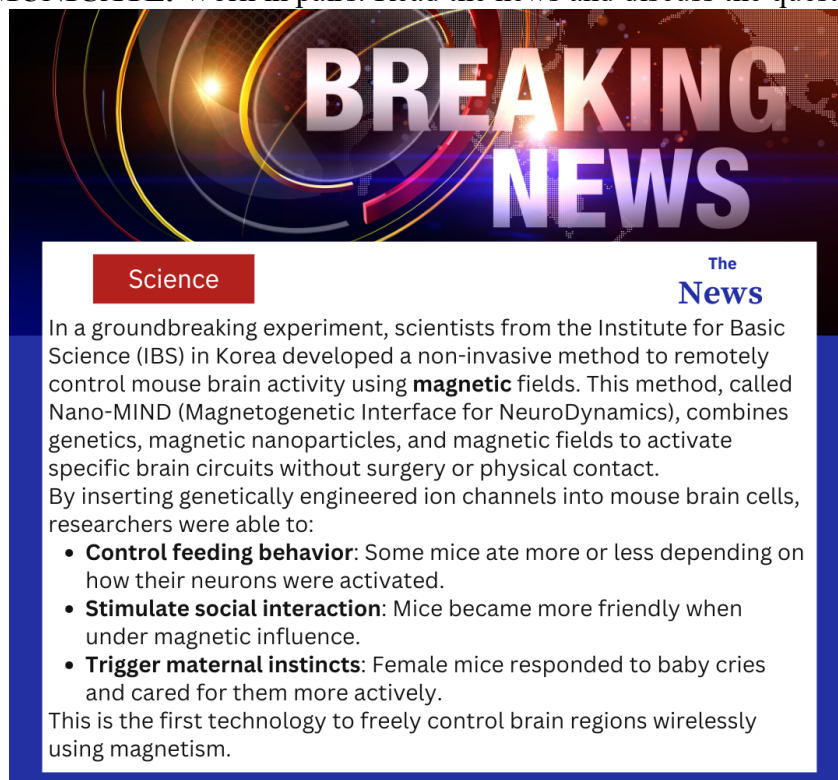
Tech Support Response: Diagnose the issue (memory triggering, software error, etc.)

Situation 8

We've been using mind-control technology for group brainstorming sessions, but lately, the system seems to sync everyone's thoughts to one central idea, causing people to have the same thoughts at the same time. This is causing groupthink and limiting creativity. Can this be fixed?

Tech Support Response: Diagnose the issue (group synchronization issue, feedback loop, etc.)

Task 62. COMMUNICATE. Work in pairs. Read the news and discuss the questions below.



BREAKING NEWS

Science **The News**

In a groundbreaking experiment, scientists from the Institute for Basic Science (IBS) in Korea developed a non-invasive method to remotely control mouse brain activity using **magnetic** fields. This method, called Nano-MIND (Magnetogenetic Interface for NeuroDynamics), combines genetics, magnetic nanoparticles, and magnetic fields to activate specific brain circuits without surgery or physical contact. By inserting genetically engineered ion channels into mouse brain cells, researchers were able to:

- **Control feeding behavior:** Some mice ate more or less depending on how their neurons were activated.
- **Stimulate social interaction:** Mice became more friendly when under magnetic influence.
- **Trigger maternal instincts:** Female mice responded to baby cries and cared for them more actively.

This is the first technology to freely control brain regions wirelessly using magnetism.

(adapted from <https://www.sciencealert.com/scientists-mind-controlled-mice-remotely-in-extraordinary-world-first>)

1. What makes Nano-MIND different from previous mind-control experiments in animals?
2. How could this technology help scientists understand the human brain better?
3. Can you think of other animals or behaviors where this technology might be tested?
4. Do you think Nano-MIND technology should be adapted for human treatment?
5. How far are we from seeing Nano-MIND-like tools used in everyday life?
6. What do you imagine life would be like if we could safely switch moods or behaviors on and off using magnets or BCI?

Task 63. COLLABORATE. Work in small groups. Discuss the hypothetical BCI game features.

Would You Play It?

- A horror game that reads your fear levels and becomes scarier the more afraid you are.
- A multiplayer game where your focus controls your speed.
- A VR game where your imagination creates the world in real time.
- A puzzle game that gets harder every time you get bored, based on your brain activity.

- An educational game that can tell when you don't understand and automatically changes the explanation or gives hints.
- A shooter game that disables your weapon if it detects anger or aggression.
- A mystery game that hides clues when you focus too hard and reveals them when you let your mind wander.
- A memory game that can erase certain parts of the story from your short-term memory – you must replay and re-learn them.
- A team-building game where you can only win by mentally syncing with your teammates.
- A horror survival game that ends instantly if it detects you've completely lost hope.
- A music rhythm game that creates songs based on your brainwaves and emotional patterns.
- A dream game you play while you sleep – your brain creates the story, and you remember it when you wake up.

1. Would you play it?
2. What are the pros and cons?
3. Could it become addictive or dangerous?

Task 64. COLLABORATE. Work in pairs. Draw a timeline of BCI in gaming from 2025 to 2050.

BCI Gaming Timeline

- What kinds of games will exist in 5, 10, 25 years?
- Can you describe one future game or feature ?
- How will it change entertainment?

Task 65. COLLABORATE. Divide into two teams to have a debate about mind-controlled games.

Some people think mind-controlled games are the future of fun and learning, while others are worried about privacy and health. What do you think? Are mind-controlled games a great innovation or a possible invasion of our minds?

Mind-Controlled Games: Innovation or Invasion?

Innovation	Invasion
1. You can control the game with your thoughts – no need for controllers.	1. Games may collect your private thoughts and emotions.
2.	2.

Task 66. COLLABORATE. Divide into two teams to have a competition. Slips of paper with questions from all sections of the unit should be placed into a box. Follow the rules:

- Each team sends a representative to draw a question and answer it within 30 seconds.
- If the team answers correctly and fast, they get 2 points.
- If they need more time (up to 1 minute), they get 1 point.
- If they give a wrong answer, the other team can steal the point by answering correctly.
- Bonus Round: Answer tricky questions at the end. Each correct answer is worth 3 points.

Team Challenge

Round 1. Brain-Computer Interfaces

1. What is a non-invasive BCI? Give an example.
2. How does a BCI translate brain signals into commands?
3. Name one current use of BCIs in medicine.
4. How are EEGs involved in BCIs?

Round 2. Neurogadgets

1. What is the main purpose of neurogadgets like smart headbands?
2. How can neurogadgets help in gaming or relaxation?
3. Give an example of a neurogadget and its function.
4. What makes a device a neurogadget rather than just a wearable?

Round 3. Mind-Control Tech Around Us

1. How is mind-control tech used in entertainment?
2. Name one ethical concern related to mind-controlled devices.
3. How can these technologies affect user privacy?
4. Which areas outside healthcare and gaming use mind-controlled tech?

Round 4. Word and Idioms.

1. Which word describes a signal in your brain or a sudden clever idea?
2. What technology is often used in brain research and scans but is not a BCI?
3. Which idiom means someone is acting crazy or irrationally?
4. Which word means the body part that helps move artificial limbs with brain signals?

Round 5. Tricky questions.

1. Can a BCI read your thoughts? Why or why not?
2. What might be the risk of using neurogadgets at home without supervision?
3. Is it ethical to use mind-controlled tech to monitor emotions at work? Why?
4. Could BCIs be hacked like regular computers? What could be the consequences?

Task 67. ANALYZE. Read the case study and discuss possible solutions to the problem.

A small startup recently launched a brain-sensing wearable that lets people open apps or type short messages on their phones just by thinking. The idea sounded futuristic and practical, especially for people with limited mobility or for professionals who are always on the go and need their hands free. In quiet settings like someone sitting at home or working alone in an office it mostly did what it was supposed to. But when people started using it out in public, it was a different story. Whether they were walking, riding the bus, or just around lots of noise and motion, the device didn't handle it well. Things became glitchy, and users noticed that it missed commands or reacted too slowly. The team behind the product thinks it might be picking up interference from body movement or all the background signals floating around in busy areas. They're now trying to figure out how to make the system more reliable when people are out and about, not just sitting still indoors.

What could be done to help brain-signal-based smartphone control work better in noisy, real-world environments where people are on the move?

WRITING

Task 68. ANALYZE. Read the opinion paragraph written by a student for a class blog. Then answer the questions that follow.

Do Brain-Computer Interfaces Belong in Everyday Life?

I believe brain-computer interfaces will become part of daily life within the next 10 to 15 years. We already use voice assistants and smart home devices, and BCIs are the next logical step. With BCIs, people with physical disabilities could control wheelchairs or home systems using only their thoughts, which would make life more independent. In education and work, BCIs might help people focus better or access information faster. Of course, there are risks, such as privacy concerns or tech failures, but I think the benefits will outweigh the drawbacks. Society just needs time to adapt and regulate this technology properly.

1. What is the author's opinion?
2. What reasons does the author give to support it?
3. What are the positive effects of BCIs mentioned?
4. What concern is mentioned?

5. Underline linking phrases.
6. Is the paragraph well-organized and convincing? Why?

Task 69. WRITE. Write an opinion paragraph on one of the following topics. Choose one and write 5–6 sentences giving your opinion. Support it with reasons and examples. Use expressions from the model to organize your ideas.

Topics:

- Will neurogadgets like brain-controlled headsets become popular among gamers and students?
- Should mind-reading technology be limited only to medical use?
- Do you think society is ready for devices that respond to our thoughts?

Useful language:

- I believe that...
- In my opinion,...
- One example is...
- This technology could help...
- However,... / On the other hand,...
- That's why I think...

LANGUAGE FOCUS

Task 70. STUDY AND ANALYZE. Look at the rule about all, every and whole, both, either, and neither, study in what situations they are used.

ALL, EVERY AND WHOLE

USE	EXAMPLES
All refers to more than two people, things, or groups. It has a positive meaning and is followed by a plural verb. It is the opposite of none.	All the devices were connected to the network. All of them received the software update. They were all excited about the new VR headset. All three prototypes passed the quality test.
All + that-clause means "the only thing" and takes a singular verb.	All that the AI system did was generate error messages.
Every is used with singular countable nouns. It refers to a group of people or things and means "all," "everyone," "everything," etc.	The antivirus software runs a scan every hour . He tested every sensor before launching the robot.
Whole (= complete) is used with singular countable nouns. We use a/the/this/my, etc. + whole + noun.	The engineer rebuilt the whole motherboard after the power surge. ALSO: He had to replace all the components inside the motherboard.
We don't use whole with uncountable nouns.	The hacker deleted all the data. (NOT: the whole data.)

BOTH, EITHER AND NEITHER

USE	EXAMPLES
Both Refers to two people, things, or groups. It has a positive meaning and is followed by a plural verb. It is the opposite of neither/not either.	Both Apple and Samsung manufacture smart watches. They are both working on AI-powered assistants. Both of the processors perform well in gaming. Both models support 5G connectivity.

Either Refers to two people, things, or groups and is followed by a singular countable noun.	You can choose either laptop ; both have excellent battery life. Either operating system is compatible with the software.
Either of + plural noun phrase can be followed by either a singular or plural verb.	Either of these two graphics cards suit(s) your gaming needs. BUT Either card is a great choice for 4K video editing.
We can use not ... either (of) instead of neither (of). Either can also be used at the end of a negative sentence.	I tested two VR headsets, but I didn't like either of them . My friend doesn't use cloud storage, and I don't either .
Neither Refers to two people, things, or groups and has a negative meaning.	Neither of the smartphones is/are waterproof. Neither of the brands offers open-source software. BUT Neither laptop has a touchscreen.
Neither ... nor / Either ... or take either a singular or plural verb, depending on the subject that follows nor or or.	Neither Intel nor AMD has released a new processor this month. Either the IT manager or the developers are responsible for fixing the bug.

Task 71. PRACTICE. Circle the correct item.

1. **Neither / Either** Neuralink nor NextMind has released a fully commercial brain implant yet.
2. **Both / Every** neurogaming headsets track brain activity in real time.
3. He tested the **all / whole** neurofeedback system before using it on patients.
4. We monitored brain activity during the experiment **both / all** days to collect enough data.
5. The AI-based algorithm processed the **whole / all** the EEG recordings in under a minute.
6. **Either / Neither** of the neuroprosthetic arms is compatible with the new software update.
7. **Both / Every** neurostimulation techniques showed promising results in pain management.
8. The scientist didn't detect strong activity in **either/neither** brain region.
9. He spent the **all / whole** night adjusting the settings of the brainwave-controlled drone.
10. **Either / Neither** the neurogaming headset nor the eye-tracking device responded correctly during testing.
11. **Every / All** neurostimulator device must be tested for safety before clinical use.
12. The patient can wear the neuroheadset on **either /neither** side of the head.

Task 72. PRACTICE. Rewrite the sentences using both ... and, neither ... nor, or either ... or.

1. The EEG headset can connect to a smartphone, or the VR headset can.
2. Brain stimulation doesn't work for Lisa. It doesn't work for Tom, either.
3. Neuralink is working on a brain implant. Synchron is working on one, too.
4. Scientists might use AI to analyze brain waves, or they might use machine learning.
5. This neurofeedback game doesn't improve reaction speed. That one doesn't either.
6. The neurostimulator helps with pain relief. It also helps with muscle recovery.
7. If the EEG cap doesn't detect signals correctly, the software will have to be adjusted.
8. Patients don't feel discomfort when using the brain implant. They don't feel any pain, either.
9. Neurogaming can improve concentration. It can also enhance memory.
10. The AI-powered brain scanner needs calibration. So does the EEG headset.

Task 73. PRACTICE. Fill in all, every, both, either, neither.

Tech Support: Hi! Need help with your neurogaming headset?

User: Yeah, I'm trying to connect 1) _____ the headset and the neural controller, but it's not working.

Tech Support: Gotcha. Let's check. (A few seconds later) Hmm... 2) _____ device should be showing up, but I don't see them paired. Are the lights on?

User: Nope. 3) _____ of them are responding. They're 4) _____ just stuck on standby.

Tech Support: Ah, classic. Want me to reset them?

User: Yeah, please. It's so annoying. 5) _____ my neurogadgets are supposed to sync, but none of them are working together.

Tech Support: I hear you. 6) _____ time this happens, it's usually a firmware thing. Let me check for updates.

User: That makes sense.

Tech Support: Yup! 7) _____ our latest headsets need a quick software patch. We've got one for 8) _____ model. Once that's done, you should be good to go!